CS 3015 - Robotics

Semester Project – Requirement Document

Faculty: Umarah Qaseem

You are required to **design** and **implement** a robotics semester project. It must have hardware and software components integrated with each other. The project title/topic is open ended, but it must come inside the robotics domain. For this, a session was conducted in class to brainstorm the ideas and group discussion. The approved list of projects is as follows. To change your selected idea, you must get it approved by your teacher first.

List of approved projects:

1- **Multi-functional RC Robotic Car**
2- **Excavator Arm**
3- **Precision Farming Robot for weeds extraction**
4- **A synchronized Swarm robotics group using nano robots for task division and working in unison**
5- **AI powered robotic system designed to automate stitching tasks**
6- **Aerial Robots for delivering goods and services during natural disasters**
7- **Cricket Ball Speed gun**

Code and Modularity:

The code should follow the programming conventions, and it should be modular. Everything should be well referenced if it is taken from any source. You can use any language for programming (Arduino programming language, micro python, MATLAB etc)

Presentation:

The overall hardware presentation carries marks. GUI is optional, but marks will be given based on efforts.

Efforts:

Efforts count the most!

Plots and Visualization:

If there are any plots, please do put them in your report.

Rules and Guidelines:

1. This is a group assignment.
2. You are responsible for forming your groups.
3. No excuses will be accepted for incomplete contributions.
4. The assignment will be evaluated as a whole; therefore, the division of work is your responsibility.
5. Each group member is expected to contribute actively. Failure to do so may result in penalties or exclusion from the evaluation.
6. Ensure that all group members are familiar with and agree upon the division of tasks.
7. Group disputes will not be entertained. It is the group's responsibility to resolve any internal issues.

Report:

A word document (converted to pdf later) in **proper** report format. Learn how to present your efforts! It should contain everything in a structured way. Complete Code should be present at the very end of your report. Although you can put screen shots of a function or two which is unique or if you want to highlight something. Write any functions or features that you have added in the report as well. Also mention the work division between the group members in the report.

Deliverables:

1- Code file (Name it as Rollnumber_firstname_ lastname_code)
2- Report (Name it as Rollnumber_firstname_lastname_report)
3- Hardware demo in class.

Put both these in a zip file and submit on GCR. Name it as (Rollnumber_firstname_lastname_assignment1). Handwritten material (if any) must be scanned and put in report.

Evaluation:

Through report and demo

Group Members:

4 group members are allowed. (Every member will not get the same marks therefore please do your part properly)

Plagiarism:

STRICT POLICY ON CHEATING THROUGH ANY SOURCE WITHOUT REFERENCE and UNDERSTANDING. ZERO MARKS WILL BE ASSIGNED IN ALL ASSIGNMENTS AND QUIZZES.

Reports will be checked through plagiarism detector, and marks will be deducted upon the usage of generative AI for reports.

Announcement Date:

30th October 2024

Deadline:

Hardware: 2nd December 2024 In Class

Software submission 3rd December 11:50 PM